

Kripan K.C.

Geospatial Data Scientist | Remote Sensing & Earth Observation | Climate Risk Modeling

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PROFESSIONAL SUMMARY

Geospatial Data Scientist and Remote Sensing Engineer pursuing an M.Sc. in Environmental Engineering at the Technical University of Munich (TUM), specializing in Environmental Hazards/Risk and Water Resources Management. Combines applied deep learning (PyTorch, semantic segmentation, U-Net/ResNet-34) with geospatial engineering (ArcGIS, QGIS, Google Earth Engine, GDAL/Rasterio/GeoPandas) to build production-grade Earth observation and climate-risk analytics pipelines. Industry experience at Munich Re (NatCat underwriting geodata workflows) and research experience at the German Aerospace Center (DLR) and TUM, spanning flood hazard modeling, hydrology, and large-scale satellite data processing on HPC infrastructure (NVIDIA A100 GPUs).

TECHNICAL SKILLS

Programming & ML: Python, PyTorch (CUDA, GPU-accelerated training), Scikit-learn, Deep Learning (CNNs, Semantic Segmentation, U-Net, ResNet-34), Random Forest, XGBoost, SVM

Geospatial Engineering: ArcGIS Pro/Enterprise, QGIS, GDAL, Rasterio, GeoPandas, Spatial Indexing, Raster & Vector Processing

Earth Observation & Remote Sensing: Google Earth Engine, Sentinel/Landsat Imagery, Satellite Image Processing, Change Detection, EO Pipelines

Data Engineering & APIs: SOAP/REST API Integration, Multithreaded Batch Processing, Cloud Platforms, Databricks, Parallel Processing

Hydrology & Risk Modeling: HBV, HEC-RAS, HEC-HMS, Flood & Natural Hazard Modeling, Spatial Risk Analysis, Climate Risk Analytics

PROFESSIONAL EXPERIENCE

Master's Thesis Researcher

May 2026 – Present

German Aerospace Center (DLR), Earth Observation, Munich

- Designing and training a U-Net semantic segmentation model (ResNet-34 encoder, PyTorch) to classify sediment, shallow water, vegetation, and shadow in high-resolution Alpine river orthophotos.
- Benchmarking deep learning performance against a Random Forest baseline and evaluating zero-shot foundation models (e.g. SAM), identifying their limitations for continuous semantic transitions in complex river environments.
- Running large-scale training and inference on the Terrabyte HPC cluster (NVIDIA A100, 80GB VRAM; PyTorch/CUDA 12.4), with optimized dataloaders for high-resolution orthophoto patches.
- Mitigating geospatial ML pitfalls — spatial autocorrelation, inter-patch data leakage, and class imbalance — through targeted spatial splitting, sampling, and augmentation strategies. Expected thesis completion: Oct 2026.

Intern, Corporate Underwriting

Sep 2025 – Mar 2026

Munich Re, Munich

- Designed and implemented workflows for large-scale, complex geodata using hybrid raster/vector algorithms, spatial indexing, and parallelized, cloud-based execution.
- Developed and refactored a GUI-based geodata enrichment tool with multithreaded batch processing and SOAP/REST API integrations within a modular Python architecture.
- Built spatial data enrichment workflows combining internal geodata with census and administrative datasets to support natural hazard and risk modeling.
- Published ArcGIS Enterprise maps and geoprocessing services for underwriting web applications; designed GIS toolboxes and frameworks for spatial risk analysis.

- Performance rated “sehr gut” (1/1, the highest grade) in the final evaluation, recognized for consistently high initiative, reliability, and contributions exceeding intern expectations.

Research Assistant

Nov 2024 – Present

Technical University of Munich (TUM), Munich

- Conducting flood risk analysis using GIS, remote sensing, and hydrological modeling tools (HEC-RAS, HEC-HMS) to support watershed-scale hazard assessment.
- Supporting environmental data analysis and research deliverables for ongoing flood and natural hazard modeling projects.
- Serving as Teaching Assistant for graduate-level courses in Flood Risk Management and Urban Hydrology, communicating technical hydrological concepts to non-specialist audiences.

Freelance GIS Analyst

Apr 2020 – Mar 2024

Upwork

- Delivered geospatial analysis and Earth observation projects for international clients, including Google Earth Engine workflows for evapotranspiration and land-surface analysis.
- Optimized global geospatial processing pipelines, reducing runtimes from days to hours.

EDUCATION

M.Sc. Environmental Engineering

CGPA: 1.4 (1.0 = highest) | Expected: Oct 2026

Technical University of Munich (TUM), Germany — Environmental Hazards/Risk & Water Resources Management

Master's Thesis (DLR): Semantic segmentation of Alpine river sediment dynamics using deep learning (U-Net, ResNet-34 encoder).

B.Tech. Environmental Engineering

CGPA: 1.3 (1.0 = highest)

Kathmandu University, Nepal

ADDITIONAL PROJECTS

- Reduced precipitation interpolation error by 45.4% using DEM-based smoothing and stratified kriging for rainfall-runoff modeling.
- Implemented ML-based glacier mapping workflows using Landsat and Sentinel-2 imagery to support climate-change analysis.

AWARDS & PRESENTATIONS

- Deutschlandstipendium Scholarship (2024/25 and 2025/26).
- Poster presentations at the International Mountain Conference (Nepal) and Technical University of Munich (TUM).

LANGUAGES

English (Professional) | German (A1–A2, Basic) | Nepali (Native)